

Controller-Mediated Intervention Cycles in Inverter-Dominated Power Grids: Self-Energy Hybridization, Resonant Topology Sensitivity, and Non-Composable Corrective Actions

Jorge L. Mayorga Taborda

Abstract—Grid-strength, damping, and inverter-control studies are often organized around individual buses, individual lines, or individual controller settings. This paper develops a different viewpoint: in inverter-dominated grids, the relevant planning object can be the *interaction among corrective actions*. Starting from an exact Schur self-energy reduction of the full linearization, a frequency-dependent network operator is constructed that retains condensed controller dynamics rather than replacing them by a scalar nodal damping coefficient. Three layers are then developed. First, a network-controller hybridization model quantifies when electrical modes and controller modes cease to be separable and introduces a scale-invariant biorthogonal family measure. Second, an exact nonlinear-eigenvalue sensitivity shows that topology acts through structural, power-flow-mediated, and controller-mediated channels; near hybrid-mode coalescence the topology sensitivity is amplified by eigenvalue conditioning. Third, finite-rank interventions are lifted to an action space. The resulting intervention Green matrix yields an exact perturbation determinant, an acyclic composability theorem, a strongly-connected-component factorization, and an all-order closed-cycle expansion with a truncation bound. These results imply that individually beneficial interventions need not compose beneficially.

Mechanism-level IEEE 39-bus experiments support the theory. A controller-resonant line exhibits a sign reversal from a network-only damping sensitivity of $+1.34 \times 10^{-4}$ to a controller-aware sensitivity of -4.27×10^{-3} . Near network-controller coalescence, the maximum line sensitivity is about $50\times$ the detuned value. Across 17,664 line/controller action pairs, a spectral interaction measure correlates with damping non-additivity (Spearman 0.673, AUC 0.897 for the strongest 5%). In a representative case, a line reinforcement improves damping by 1.45% and a controller retuning improves it by 5.39%, yet their joint application reduces the damping margin by 10.76%. A second campaign establishes exact portfolio composability results and shows that when inertia and synchronous damping are retired together, pair and triple action-cycle magnitudes scale approximately as α^{-1} and $\alpha^{-3/2}$, whereas preserving absolute damping reverses the trend. The framework therefore reframes dynamic-security planning from isolated-action ranking to controller-aware portfolio interaction analysis.

Index Terms—inverter-based resources, grid-forming converters, self-energy, nonlinear eigenvalue problem, modal interaction, topology sensitivity, dynamic security, Green functions, low-rank perturbations, intervention portfolios, higher-order interactions.

J. L. Mayorga Taborda is with Universidad de los Andes, Bogota, Colombia. This manuscript consolidates a theory-development campaign performed in August 2026. Results identified as “mechanism-level” use a topology-grounded IEEE 39-bus reduced/controller model; full manufacturer-grade GFL/GFM and EMT validation remains a required next step before universal industrial claims.

I. INTRODUCTION

INVERTER-dominated power systems create a modeling dilemma. Detailed GFL/GFM models contain PLLs, current loops, voltage loops, filters, droop/VSM dynamics, and device-specific auxiliary states; planning studies, in contrast, need interpretable and computationally reusable models. A common reduction is to collapse these details into a fixed $L-M-D$ model. That reduction is useful only when the omitted control dynamics do not change the relevant mode family, the topology sensitivity, or the ranking of candidate corrective actions.

Existing literature already covers several adjacent problems. Virtual inertia placement and primary-control allocation have been formulated using system-theoretic and perturbation methods [3], [4]. Input-output analysis has been used to study inter-area oscillations [5]. Open-loop modal resonance has long been recognized as a mechanism for strong controller-network interaction in converter-connected systems [6]–[8]. Recent work has developed graph-theoretic stability criteria for IBR grids [9], topology/operating-point stability sensitivities [11], perturbation-space interaction sensitivities [12], and robust certificates for reduced phasor-model trustworthiness [10]. These results make it difficult to claim novelty from another weak-bus score, another eigenvalue sensitivity, or the generic observation that topology and controllers interact.

This work instead focuses on a narrower but more consequential question:

When do otherwise beneficial corrective actions become non-composable because controller dynamics and network topology close a dynamic feedback cycle?

The proposed answer connects exact model reduction, modal hybridization, topology sensitivity, and finite-rank portfolio theory.

The main contributions are summarized as follows.

- 1) An exact controller-aware self-energy bridge is used as the master reduced operator. A biorthogonal network/control family fraction is proposed for this bridge; unlike a norm-based state fraction, it is invariant to diagonal state-coordinate scaling.

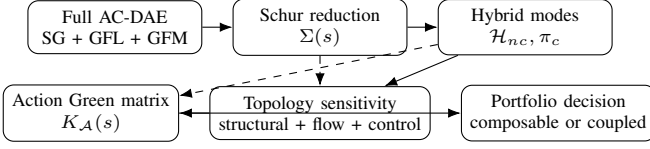


Fig. 1. Unified viewpoint. The self-energy bridge preserves condensed controller dynamics. Network–controller hybridization modifies topology sensitivity, while the same controller-aware operator induces action-space feedback cycles that determine portfolio composability.

- 2) An exact nonlinear-eigenvalue sensitivity is specialized to topology actions. It separates structural, flow-mediated, and controller-mediated contributions. A two-mode hybridization model explains sensitivity amplification near network–controller coalescence.
- 3) Corrective actions are embedded in a finite-dimensional intervention space. The resulting action Green matrix yields an exact perturbation determinant, an acyclic composability theorem, an SCC factorization, and an all-order action-cycle expansion with a rigorous truncation bound.
- 4) A low-inertia/damping scaling law is established for intervention cycles. Under uniform co-retirement $M, D \mapsto \alpha(M, D)$, isolated-mode pair and triple cycle kernels scale as α^{-1} and $\alpha^{-3/2}$. If inertia decreases while absolute damping is preserved, the signs of the exponents reverse.
- 5) IEEE 39-bus mechanism experiments identify controller-resonant topology-sensitivity reversals, near-coalescence sensitivity amplification, destructive line–controller action combinations, and predictive action-interaction scores. An industrial planning example shows why independent screening can recommend a jointly harmful investment portfolio.

The algebraic tools used here—Schur complements, nonlinear eigenvalue sensitivity, determinant lemmas, Woodbury identities, and matrix logarithm expansions—are established mathematics. The candidate originality lies in their synthesis into a controller-aware *intervention-cycle* theory, the composability results, the inertia–damping interaction scaling, and the demonstrated decision consequences. A final literature audit and full SG/GFL/GFM benchmark remain necessary before priority claims are made.

II. CONTROLLER-AWARE DYNAMIC OPERATOR

A. Exact Schur self-energy

Consider a linearized dynamic model after algebraic elimination,

$$\begin{bmatrix} \dot{x}_r \\ \dot{x}_c \end{bmatrix} = \begin{bmatrix} A_{rr} & A_{rc} \\ A_{cr} & A_{cc} \end{bmatrix} \begin{bmatrix} x_r \\ x_c \end{bmatrix} + \begin{bmatrix} B_r \\ B_c \end{bmatrix} u, \quad (1)$$

where x_r denotes retained network/interface coordinates and x_c denotes controller and auxiliary states. For $s \notin \text{spec}(A_{cc})$,

$$x_c = (sI - A_{cc})^{-1}(A_{cr}x_r + B_c u). \quad (2)$$

Substitution gives

$$S_r(s)x_r = B_{\text{eff}}(s)u, \quad (3)$$

with

$$S_r(s) = sI - A_{rr} - \Sigma_A(s), \quad (4)$$

$$\Sigma_A(s) = A_{rc}(sI - A_{cc})^{-1}A_{cr}. \quad (5)$$

When the retained coordinates admit a second-order electromechanical representation, we write

$$T(s) = s^2 M + s \Sigma(s) + L, \quad (6)$$

where $\Sigma(s)$ is generally matrix-valued, non-diagonal, and frequency dependent.

Proposition 1 (Exact retained-space equivalence). *Except for controller poles treated explicitly in the Schur bookkeeping, zeros of $\det S_r(s)$ coincide with full-system eigenvalues whose modes have nonzero retained-space participation.*

Proof. Taking the determinant of $sI - A$ blockwise gives

$$\det(sI - A) = \det(sI - A_{cc}) \det S_r(s), \quad (7)$$

whenever $sI - A_{cc}$ is invertible. Analytic continuation extends the equality away from isolated singular bookkeeping points. Hence the full determinant factorizes into internal-controller poles and the Schur-retained nonlinear eigenvalue problem. $\square$

For a diagonalizable A_{cc} ,

$$(sI - A_{cc})^{-1} = \sum_c \frac{r_c l_c^*}{s - \lambda_c}, \quad (8)$$

so the controller self-energy has the pole-residue expansion

$$\Sigma(s) = D_0 + \sum_c \frac{R_c}{s - \lambda_c}, \quad R_c = C r_c l_c^* B. \quad (9)$$

Thus each controller mode produces a rational spectral footprint whose pole location and matrix residue separately encode frequency and spatial reach.

B. Scale-invariant mode-family participation

A norm-based fraction

$$\pi_c^{\text{norm}} = \frac{\|x_c\|^2}{\|x_r\|^2 + \|x_c\|^2} \quad (10)$$

changes under state-unit rescaling. For right and left eigenvectors r, l , define instead

$$\pi_c^{\text{bio}} = \frac{\sum_{i \in \mathcal{C}} |\bar{l}_i r_i|}{\sum_i |\bar{l}_i r_i|}. \quad (11)$$

Proposition 2 (Diagonal-scale invariance). *Under an invertible diagonal state transformation $x' = Sx$, π_c^{bio} is invariant.*

Proof. The transformed right and left eigenvectors satisfy $r' = S r$ and $l' = S^{-H} l$. For each coordinate, $\bar{l}'_i r'_i = \bar{l}_i r_i$, so every term in (11) is unchanged. $\square$

Figure 2 verifies the issue numerically: a controller-state scaling from 10^{-6} to 10^6 drives the norm-based fraction from approximately zero to one while the biorthogonal fraction remains 0.597033.

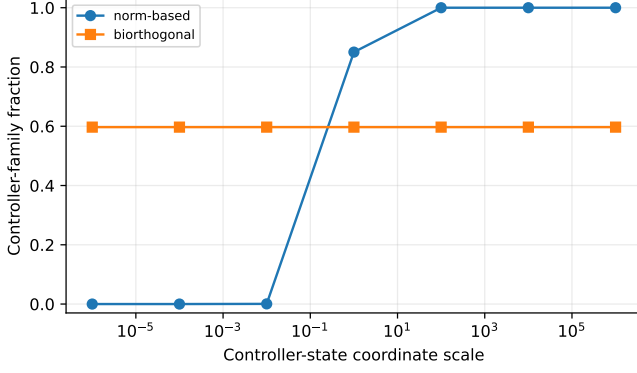


Fig. 2. Coordinate-scaling experiment. The norm-based controller fraction is arbitrary under state-unit changes; the biorthogonal fraction remains invariant.

III. NETWORK-CONTROLLER HYBRIDIZATION THEORY

A. Two-mode reduction

Let λ_n be an uncoupled network mode and λ_c a controller mode. A local two-mode representation is

$$H_{nc} = \begin{bmatrix} \lambda_n & g_{nc} \\ g_{cn} & \lambda_c \end{bmatrix}, \quad (12)$$

with poles

$$\lambda_{\pm} = \frac{\lambda_n + \lambda_c}{2} \pm \sqrt{\frac{(\lambda_n - \lambda_c)^2}{4} + g_{nc}g_{cn}}. \quad (13)$$

Define the dimensionless hybridization indicator

$$\mathcal{H}_{nc} = \frac{4|g_{nc}g_{cn}|}{|\lambda_n - \lambda_c|^2}. \quad (14)$$

The indicator is not proposed as a universal resonance certificate; it is a reduced diagnostic for the validity of a network/control family separation.

Figure 3 shows the mechanism-level sweep. Away from resonance, one mode is network-family and one is control-family. Around the interaction region, the family fractions mix while $\mathcal{H}_{nc}$ approaches and exceeds unity. The two-mode approximation produced median relative pole error 0.1825% and maximum error 0.8723% over the sweep.

B. Near-coalescence sensitivity

Let

$$\Delta = \lambda_n - \lambda_c, \quad \mathcal{D} = \frac{\Delta^2}{4} + g_{nc}g_{cn}. \quad (15)$$

Differentiating (13) with respect to a parameter ρ yields

$$\frac{d\lambda_{\pm}}{d\rho} = \frac{\lambda'_n + \lambda'_c}{2} \pm \frac{\Delta(\lambda'_n - \lambda'_c) + 2(g_{nc}g_{cn})'}{4\sqrt{\mathcal{D}}}. \quad (16)$$

Hence topology or controller sensitivity can become large as $|\mathcal{D}|$ becomes small, i.e., near a strongly hybridized or nearly defective pair.

A numerical search found two modes at approximately

$$\lambda_1 = -0.31322 + j4.66911, \quad (17)$$

$$\lambda_2 = -0.33041 + j4.68160, \quad (18)$$

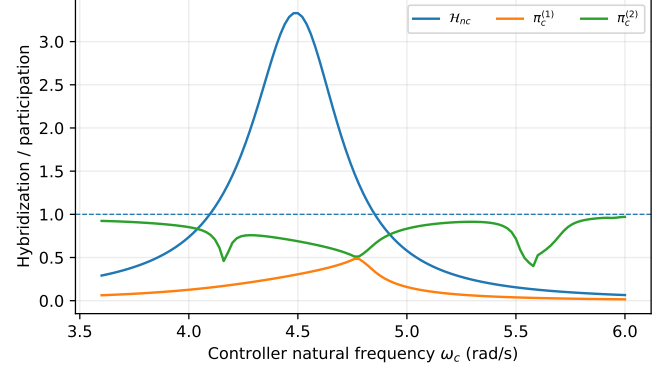


Fig. 3. Network-controller hybridization sweep. The two family fractions exchange as the controller natural frequency passes through the network modal band.

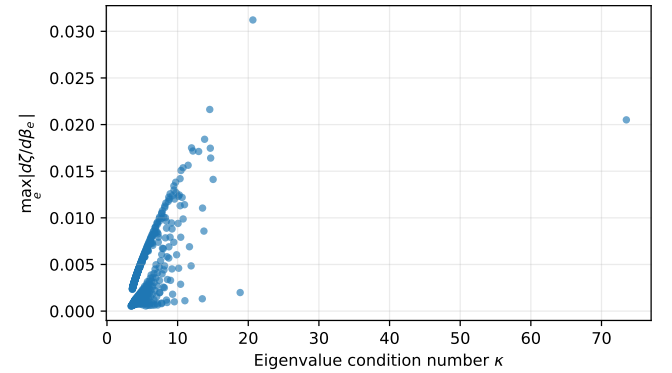


Fig. 4. Near-coalescence grid. Poor eigenvalue conditioning is associated with amplified maximum topology sensitivity in the mechanism-level controller-aware model.

with separation 0.02125, controller-family fractions 0.478 and 0.521, and eigenvalue condition numbers approximately 73.5. Near this point the maximum line-action damping sensitivity was 2.05×10^{-2} , versus 4.10×10^{-4} in a detuned case, an amplification of about 50 $\times$. Across a 527-point search, the Spearman correlation between eigenvalue condition number and maximum absolute topology sensitivity was 0.593 (Fig. 4).

IV. CONTROLLER-RESONANT TOPOLOGY SENSITIVITY

A. Exact nonlinear-eigenvalue derivative

Consider

$$\mathbf{T}(s, \rho)v = 0, \quad y^* \mathbf{T}(s, \rho) = 0, \quad (19)$$

for a simple nonlinear eigenvalue s . Differentiation gives the standard NEP sensitivity

$$\frac{ds}{d\rho} = -\frac{y^* \mathbf{T}_\rho(s, \rho)v}{y^* \mathbf{T}_s(s, \rho)v}. \quad (20)$$

For (6),

$$\frac{ds}{d\rho} = -\frac{y^*(s^2 M_\rho + s \Sigma_\rho + L_\rho)v}{y^*(2sM + \Sigma + s\Sigma_s)v}. \quad (21)$$

The denominator is the key controller-aware renormalization absent from a constant-damping QEP.

If $s = \sigma + j\omega$ and $s_\rho = a + jb$, then

$$\boxed{\frac{d\zeta}{d\rho} = \frac{\sigma\omega b - \omega^2 a}{|s|^3}}. \quad (22)$$

B. Structural, flow-mediated, and controller-mediated channels

Let the operating point x^* satisfy

$$F(x^*, z, P, Q) = 0. \quad (23)$$

If F_x is nonsingular,

$$\frac{dx^*}{dz_e} = -F_x^{-1} F_{z_e}. \quad (24)$$

For fixed M ,

$$\boxed{\frac{dT}{dz_e} = \underbrace{L_{z_e}}_{\text{structural}} - \underbrace{L_x F_x^{-1} F_{z_e}}_{\text{flow-mediated}} + s \underbrace{(\Sigma_{z_e} - \Sigma_x F_x^{-1} F_{z_e})}_{\text{controller-mediated}}}. \quad (25)$$

This decomposition explains why a local line action can have a global and controller-dependent dynamic sign.

For line 5–8 in the flow-aware mechanism study, the structural-only derivative was $+3.33 \times 10^{-7}$ while the derivative after re-solving AC power flow was -1.40×10^{-5} , demonstrating a structural/flow sign reversal.

C. Controller-resonant line reversal

A stronger example occurs for line 25–26 in the hybridized controller study. A network-only model gives

$$\left. \frac{d\zeta}{d\beta} \right|_{\text{network}} = +1.3408 \times 10^{-4}, \quad (26)$$

whereas the controller-aware self-energy model gives

$$\boxed{\left. \frac{d\zeta}{d\beta} \right|_{\Sigma} = -4.2715 \times 10^{-3}}. \quad (27)$$

The sign reverses and the magnitude increases by approximately $31.9\times$. The analytic derivative (21) agreed with finite differences to relative error on the order of 10^{-8} for well-conditioned step sizes.

We use the term *controller-resonant topology action* for an action whose network-only stiffness/damping interpretation is altered because it moves a relevant network mode into stronger interaction with a controller family. The term is descriptive; modal resonance itself is established in the literature [6]–[8].

V. DYNAMIC INTERVENTION SPECTRAL THEORY

A. Finite-rank action representation

Let the controller-aware base operator be $T_0(s)$. A portfolio $\mathcal{A} = \{a_1, \dots, a_m\}$ is represented as

$$T_{\mathcal{A}}(s) = T_0(s) + U_{\mathcal{A}}(s) C_{\mathcal{A}}(s) V_{\mathcal{A}}^*(s). \quad (28)$$

Examples include

$$\Delta T_{\text{line}} = \Delta w_e a_e a_e^T, \quad (29)$$

$$\Delta T_{D_i} = s \Delta D_i e_i e_i^T, \quad (30)$$

$$\Delta T_{M_i} = s^2 \Delta M_i e_i e_i^T, \quad (31)$$

$$\Delta T_{\text{ctrl}} = \Delta \Pi_i(s). \quad (32)$$

Define the base Green operator $G_0(s) = T_0^{-1}(s)$ and the action-space interaction operator

$$\boxed{K_{\mathcal{A}}(s) = C_{\mathcal{A}} V_{\mathcal{A}}^* G_0(s) U_{\mathcal{A}}}. \quad (33)$$

This is a finite-dimensional Birman–Schwinger-type representation of the intervention portfolio.

Theorem 1 (Intervention determinant reduction). *Whenever $T_0(s)$ is invertible,*

$$\boxed{\frac{\det T_{\mathcal{A}}(s)}{\det T_0(s)} = \det(I + K_{\mathcal{A}}(s))}. \quad (34)$$

Proof. Apply the matrix determinant lemma to (28): $\det(T_0 + UCV^*) = \det T_0 \det(I + CV^* T_0^{-1} U)$. $\square$

Thus full-grid physics is embedded in G_0 , while the spectral effect of an m -action portfolio is encoded by an $m \times m$ nonlinear matrix.

B. Exact composability factorization

Write

$$K = D + E, \quad D = \text{diag}(K), \quad E = \text{offdiag}(K), \quad (35)$$

and define

$$\boxed{Q = (I + D)^{-1} E}. \quad (36)$$

Then

$$\boxed{\det(I + K) = \left[\prod_{a \in \mathcal{A}} (1 + D_{aa}) \right] \det(I + Q)}. \quad (37)$$

The product is the individual-action contribution; $\det(I + Q)$ is the pure interaction correction.

Theorem 2 (Acyclic action composability). *If the directed graph induced by Q is acyclic, then*

$$\boxed{\det(I + Q) = 1}, \quad (38)$$

and the portfolio perturbation determinant equals the product of the individual-action factors.

Proof. Every DAG admits a topological ordering. Under the corresponding permutation, Q is strictly upper triangular. Hence $I + Q$ is triangular with unit diagonal, so its determinant is one. $\square$

Theorem 3 (Feedback-core/SCC factorization). *Let Q_c denote the diagonal blocks obtained after permuting Q into its strongly-connected-component condensation form. Then*

$$\det(I + Q) = \prod_{c \in \text{SCC}} \det(I + Q_c). \quad (39)$$

Proof. The SCC permutation makes Q block upper triangular. The determinant of a block upper triangular matrix is the product of the diagonal-block determinants. $\square$

Equations (37)–(39) imply that non-composable spectral effects are confined to action-space feedback cores, not necessarily to physically adjacent devices.

C. All-order action-cycle expansion

Theorem 4 (Closed-cycle expansion). *If $\rho(Q) < 1$,*

$$\log \det(I + Q) = \sum_{p=1}^{\infty} \frac{(-1)^{p+1}}{p} \text{tr}(Q^p). \quad (40)$$

If Q has zero diagonal, the $p = 1$ term vanishes. The $p = 2$ term sums directed two-cycles, the $p = 3$ term sums directed three-cycles, and higher terms sum closed action walks of increasing order.

Proof. For $\rho(Q) < 1$, the matrix logarithm has the convergent Neumann series $\log(I + Q) = \sum_{p \geq 1} (-1)^{p+1} Q^p / p$. Taking traces and using $\log \det = \text{tr} \log$ gives (40). $\square$

For a zero-diagonal Q ,

$$-\frac{1}{2} \text{tr} Q^2 = - \sum_{i < j} q_{ij} q_{ji}, \quad (41)$$

and

$$\frac{1}{3} \text{tr} Q^3 = \sum_{i,j,k \text{ cycles}} \frac{q_{ij} q_{jk} q_{ki}}{3}. \quad (42)$$

This provides a formal basis for pairwise and genuine three-action interaction kernels.

Theorem 5 (Truncation bound). *Let $m = |\mathcal{A}|$ and $q = \|Q\|_2 < 1$. If (40) is truncated after order p , the remainder satisfies*

$$|\mathcal{R}_p| \leq \frac{mq^{p+1}}{(p+1)(1-q)}. \quad (43)$$

Proof. Using $|\text{tr}(Q^k)| \leq m \|Q^k\|_2 \leq mq^k$,

$$|\mathcal{R}_p| \leq m \sum_{k=p+1}^{\infty} \frac{q^k}{k} \leq \frac{m}{p+1} \sum_{k=p+1}^{\infty} q^k, \quad (44)$$

which yields (43). $\square$

The theorem defines an interaction-order gate: choose the smallest p^* whose bound is below a prescribed decision tolerance. In 600 random complex interaction matrices (4800 truncations), no bound violations were observed; the maximum observed error-to-bound ratio was 0.358.

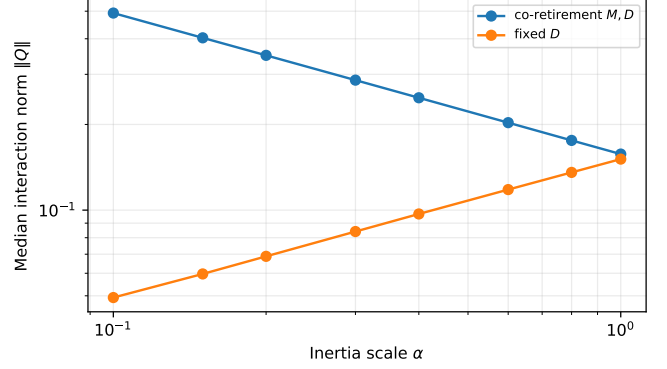


Fig. 5. Interaction-norm scaling. Co-retirement of inertia and damping amplifies action coupling as α decreases; preserving absolute damping reverses the trend.

VI. LOW-INERTIA/DAMPING SCALING OF ACTION CYCLES

A high-risk hypothesis examined in this work was that lower inertia universally increases portfolio non-composability. The experiments falsified that statement. The supported result depends jointly on inertia and damping.

Consider a single isolated network mode with modal mass m_k , damping d_k , and stiffness λ_k . Under uniform co-retirement

$$M_\alpha = \alpha M_0, \quad D_\alpha = \alpha D_0, \quad (45)$$

the natural frequency scales as

$$\omega_k(\alpha) = \sqrt{\frac{\lambda_k}{\alpha m_k}} \propto \alpha^{-1/2}. \quad (46)$$

At $s = j\omega_k$, inertia and stiffness cancel in the modal dynamic stiffness, leaving

$$|t_k(j\omega_k)| = \alpha d_k \omega_k \propto \alpha^{1/2}. \quad (47)$$

Therefore the modal Green magnitude scales as

$$|G_k(j\omega_k)| \propto \alpha^{-1/2}. \quad (48)$$

A closed action cycle of order p containing p Green factors then scales as

$$|K^{(p)}| \propto \alpha^{-p/2}. \quad (49)$$

Thus pair and triple cycles scale as α^{-1} and $\alpha^{-3/2}$.

If inertia is reduced while absolute damping is preserved, $D_\alpha = D_0$, then $|t_k| = d_k \omega_k \propto \alpha^{-1/2}$, so

$$|G_k| \propto \alpha^{+1/2}, \quad |K^{(p)}| \propto \alpha^{+p/2}. \quad (50)$$

Hence the direction of the composability trend reverses.

Figure 5 confirms the opposite trends. In 160 random low-damping isolated-mode networks, measured co-retirement slopes were -0.989 (pair), -1.469 (triple), and -0.496 ($\|Q\|$), with $R^2 > 0.99989$. With fixed damping, the slopes were $+0.956$, $+1.440$, and $+0.488$.

The interaction-order consequence is shown in Fig. 7. In the co-retirement regime, the required certified order increases as α falls, while fixed damping reduces the required order.

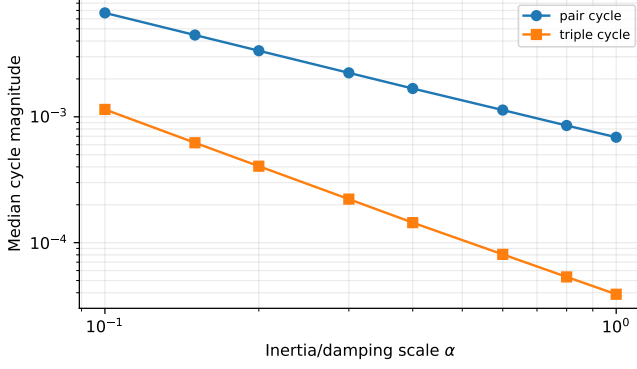


Fig. 6. Pair and triple action-cycle magnitudes under co-retirement. Measured slopes match the isolated-mode predictions -1 and $-3/2$.

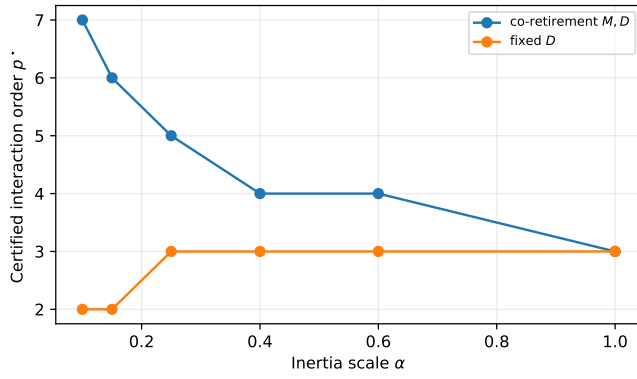


Fig. 7. Certified interaction order p^* versus inertia scale for a representative action strength. The relevant mechanism is loss of inertia *and* dissipative support, not inertia alone.

VII. IEEE 39-BUS MECHANISM EXPERIMENTS

A. Setup and reproducibility boundary

The electrical topology and solved operating point are based on the MATPOWER IEEE 39-bus New England test case [16]. Generator buses 30–39 are retained in the electromechanical reduced model after Kron reduction; controller states are added explicitly for the mechanism experiments. The controller studies use a second-order oscillatory controller coupled to a selected bus,

$$\dot{z}_1 = z_2, \quad (51)$$

$$\dot{z}_2 = -\omega_c^2 z_1 - 2\zeta_c \omega_c z_2 + b\omega_i, \quad (52)$$

$$q_i = k z_1. \quad (53)$$

Its exact self-energy contribution is

$$s\Sigma_c(s) = \frac{skb}{s^2 + 2\zeta_c \omega_c s + \omega_c^2} e_i e_i^T. \quad (54)$$

The model is deliberately transparent and exposes a complex controller pole pair. It is not claimed to reproduce a specific commercial PLL or GFM implementation.

The Schur/full-state determinant identity was checked at 100 random complex points; the maximum relative error was 5.8×10^{-15} .

TABLE I
NETWORK-CONTROLLER HYBRIDIZATION RESULTS

Test	Result
Schur/full determinant identity	max rel. error 5.8×10^{-15}
Two-mode pole reduction	median error 0.1825%, max 0.8723%
Strong hybridization point	$\mathcal{H}_{nc} = 1.119, \pi_c \approx 0.398/0.597$
Near-coalescence pair	separation 0.02125, $\kappa \approx 73.5$
Max line sensitivity near/far	$2.05 \times 10^{-2} / 4.10 \times 10^{-4}$
Conditioning-sensitivity correlation	Spearman 0.593 over 527 points
Controller-resonant line 25–26	$+1.34 \times 10^{-4} \rightarrow -4.27 \times 10^{-3}$

B. Hybridization and line-sensitivity results

Table I summarizes the principal tests.

C. Controller-mediated mixed-action cycles

For every operating/controller point, line reinforcements and controller retunings were evaluated both separately and jointly. Define the mixed finite-action non-additivity in the damping margin as

$$\mu_{ec} = \zeta_{ec} - \zeta_e - \zeta_c + \zeta_0. \quad (55)$$

A spectral interaction observable was formed directly from the exact dynamic-stiffness determinants,

$$\mathcal{I}_{ec}(s) = \log |\det \mathbf{T}_{ec}(s)| + \log |\det \mathbf{T}_0(s)| - \log |\det \mathbf{T}_e(s)| - \log |\det \mathbf{T}_c(s)|. \quad (56)$$

Across 17,664 line/controller pairs, $|\mathcal{I}_{ec}|$ had Spearman correlation 0.673 with $|\mu_{ec}|$ and AUC 0.897 for identifying the top 5% strongest non-additivities. Across a separate 23,000 line-GFM-like pair campaign, an analogous interaction score achieved Spearman 0.747 and AUC 0.894.

The hybridization index itself was strongly associated with the maximum mixed-action interaction across operating points: Spearman 0.920. This supports the causal chain

$$\text{hybridization} \rightarrow \text{large action coupling} \rightarrow \text{portfolio non-composability}. \quad (57)$$

D. Flagship destructive portfolio

A representative case has baseline damping margin

$$\zeta_0 = 0.045218. \quad (58)$$

Reinforcing line 26–29 gives

$$\zeta_e = 0.045872 \quad (+1.45\%), \quad (59)$$

and retuning the controller natural frequency from 5.1 to 4.8 rad/s gives

$$\zeta_c = 0.047657 \quad (+5.39\%). \quad (60)$$

Yet the joint action gives

$$\zeta_{ec} = 0.040352 \quad (-10.76\%). \quad (61)$$

The interaction term is $\mu_{ec} = -7.96 \times 10^{-3}$. The hybridization indicator increases from 0.381 (baseline) to 0.428 (line), 0.982

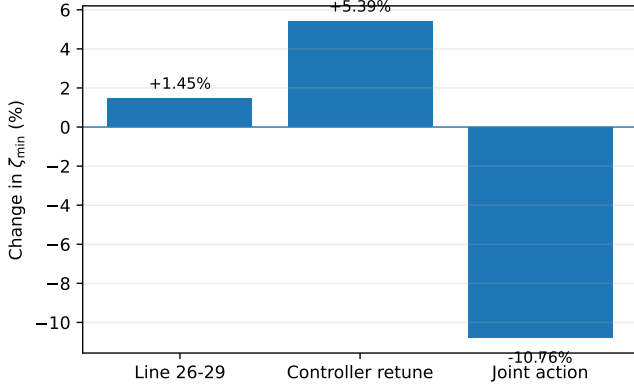


Fig. 8. Industrial planning counterexample at mechanism level. Each action is damping-positive when screened alone; the combined portfolio is damping-negative relative to baseline.

(retuning), and 1.121 jointly. Figure 8 visualizes the decision consequence.

The same qualitative phenomenon was observed in randomized high-IBR campaigns. With moderate action sizes, 11.5% of 200 scenarios contained at least one destructive line–GFM-like pair; under a stressed action campaign, the fraction was 23.5%. These are not asserted as universal frequencies; they establish that the phenomenon is not restricted to a single hand-tuned case.

VIII. LINE–LINE AND HIGHER-ORDER PORTFOLIO INTERFERENCE

The controller-mediated result is consistent with a broader portfolio phenomenon already observed for topology-only interventions. In one high-IBR case, line 19–20 alone improved $\zeta_{\min}$ by 15.8% and line 20–34 alone by 9.1%, but their joint reinforcement reduced $\zeta_{\min}$ by 7.55% below baseline. The three highest-ranked individual line actions together reduced the margin by 6.87%, whereas an interaction-aware triple improved it by 19.59%.

For the set function $F(S) = \zeta_{\min}(S)$, the pair Möbius interaction is

$$\mu_{ef} = F(ef) - F(e) - F(f) + F(\emptyset). \quad (62)$$

Both positive and negative μ_{ef} were observed, together with violations of monotonicity. Hence this damping-margin objective is, in the tested regime, neither universally monotone, submodular, nor supermodular. This matters because isolated-action or greedy guarantees from other actuator-selection objectives cannot be imported without verifying the relevant set-function structure.

For pairwise line portfolios, the Green interaction kernel

$$K_{ef} = |\beta_e \beta_f r_{ef} r_{fe}| \quad (63)$$

had Spearman correlation 0.772 with $|\mu_{ef}|$ and AUC about 0.96 for detecting the strongest 5% interactions. For 286 triples, the three-cycle kernel

$$K_{efg}^{(3)} = |\beta_e \beta_f \beta_g (r_{ef} r_{fg} r_{ge} + r_{eg} r_{gf} r_{fe})| \quad (64)$$

had Spearman correlation 0.874 with the genuine third-order Möbius interaction. These are the finite-action manifestations of the cycle expansion (40).

IX. INDUSTRIAL INTERPRETATION AND PLANNING WORKFLOW

The immediate industrial use is not to replace detailed EMT studies; it is to avoid screening candidate upgrades as if they were independent.

Consider a transmission operator evaluating a line reinforcement and a GFM/controller retuning. Conventional isolated screening produces two positive business cases: the line increases damping and the retuning increases damping. In the flagship example, executing both investments would instead reduce the damping margin by more than 10%. The mechanism is not visible in isolated action values: the line shifts the network mode, the controller retune shifts the controller family, and the pair closes a strong mixed action cycle near a hybridization region.

A controller-aware workflow is therefore:

- 1) Solve the AC operating point and linearize the full SG/GFL/GFM model.
- 2) Form a Schur self-energy for retained network/interface states and classify the margin-setting mode using scale-invariant biorthogonal participation.
- 3) Compute controller-aware topology sensitivities using (21) and, where topology changes the operating point, the three-channel derivative (25).
- 4) Build K_A for the candidate interventions. If the action graph is acyclic, the perturbation determinant is exactly composable. Otherwise, decompose into SCC feedback cores.
- 5) Use the truncation bound (43) to decide whether individual, pairwise, or higher-order interactions are required. Strongly coupled SCCs are optimized jointly.
- 6) Validate only the final candidate portfolios using the complete small-signal model and nonlinear/EMT simulations, including current limits, protections, voltage constraints, and N-1 feasibility.

The framework is relevant to at least four industry decisions: (i) transmission reinforcement coordinated with GFM/BESS deployment, (ii) controller retuning after topology changes, (iii) SG retirement/IBR replacement studies in which synchronous damping is lost together with inertia, and (iv) prioritization of expensive EMT studies toward portfolios whose action-space feedback cores are strongest.

X. ORIGINALITY BOUNDARY AND CLAIMS

Table II separates established mathematics from candidate project contributions. This distinction is necessary because several ingredients have deep prior literature.

A novelty audit through August 2026 found closely related work on graph-theoretic IBR stability [9], perturbation-driven interaction sensitivity [12], topology/operating-point sensitivity [11], and reduced-model trust certification [10]. The claim made here is narrower: the full chain from controller self-energy hybridization to resonant topology-sensitivity reversal

TABLE II
CLAIM CALIBRATION AND EVIDENCE STATUS

Item	Status in literature / this work	Evidence in this manuscript	Claim level
Schur complement, resolvent, NEP sensitivity, determinant lemma, Woodbury	Established mathematical tools	Exact identities and numerical implementation checks	Not claimed as new mathematics
Open-loop/modal resonance between converter and network modes	Established power-systems mechanism [6]–[8]	Reproduced in self-energy hybridization sweep	Used as physical foundation
Biorthogonal controller-family fraction inside self-energy workflow	Biorthogonal participation is established; application here corrects a coordinate-scaling defect of a norm fraction	Invariant over controller-state scalings 10^{-6} – 10^6	Methodological contribution
Three-channel topology derivative (structural + flow + controller)	Chain rule/implicit differentiation individually known; unified controller-aware use is candidate contribution	Structural/flow reversal observed; controller channel awaits full device-dependent Σ_x	Candidate original synthesis
Controller-resonant topology reversal	Nearby literature studies controller/network resonance and stability sensitivity separately	Line 25–26 sign reversal; analytic/FD agreement	Original experimental/planning claim candidate
Action Green/Birman–Schwinger intervention operator	Birman–Schwinger and finite-rank perturbation theory are established	Exact intervention determinant reduction	Candidate new power-systems formulation
Acyclic composability and SCC feedback-core factorization	Consequences of triangular determinant structure; not located in surveyed grid-planning literature	500 random tests; numerical error $\leq 2.22 \times 10^{-16}$	Candidate theorem for intervention portfolios
All-order action-cycle expansion + order bound	Matrix-log expansion is established; intervention interpretation and gate are candidate contributions	4800 truncations, zero bound violations	Candidate theoretical contribution
Low-inertia/damping cycle scaling	Derived for isolated-mode intervention Green factors	Measured slopes closely match $-1, -3/2, -1/2$ and reverse with fixed damping	Candidate original scaling law
Destructive line–controller portfolio	Not a generic claim that “controllers interact”; claim is isolated benefits but joint damping loss predicted by interaction geometry	+1.45%, +5.39%, jointly -10.76% ; 17,664-pair statistics	Strong mechanism-level original result candidate

and then to higher-order intervention-cycle non-composability was not found as a single planning framework in that audit. This statement should be re-audited before submission.

XI. LIMITATIONS AND REQUIRED FULL-MODEL VALIDATION

The strongest numerical claims in this manuscript are mechanism-level. They use the real IEEE 39-bus topology and operating-point data but deliberately transparent controller dynamics. They do not yet establish that a commercial GFL/GFM implementation will exhibit the same frequencies or occurrence rates. In addition:

- destructive-pair percentages depend on controller parameters and action magnitudes;
- $\rho(Q) > 1$ is a loss of the standard cycle-expansion convergence condition, not a stability boundary;
- the isolated-mode scaling laws require modal dominance and fixed action normalization;
- controller-dependent topology derivatives require a physically documented map from topology/operating point to $\Sigma(s)$;
- protection, saturation, current limits, reactive/voltage dynamics, and N-1 feasibility must be checked in full nonlinear models.

The decisive next benchmark is the full IEEE 39-bus ANDES model with documented SG/GFL/GFM controllers. The specific falsification tests are: (i) whether a controller-aware line sensitivity changes sign relative to an *LMD* reduction; (ii) whether near-hybridized full modes exhibit materially larger topology sensitivity; (iii) whether individually beneficial topology and controller actions can be jointly detrimental; and (iv)

whether the action-interaction determinant predicts the non-additive full-model damping response. Failure of these tests would appropriately downgrade the corresponding claims from general planning results to mechanism-level counterexamples.

XII. CONCLUSION

This paper develops a controller-aware theory in which dynamic-security planning is governed not only by the properties of buses and lines but also by feedback cycles among corrective actions. Exact Schur condensation converts omitted converter dynamics into a frequency-dependent self-energy. Network–controller hybridization can then amplify topology sensitivity and reverse the sign of a line’s damping effect. Lifting finite-rank interventions into action space produces an exact perturbation determinant and reveals when individual action values are composable. Acyclic action graphs compose exactly, while strongly connected feedback cores generate pairwise and higher-order interaction cycles. Under co-retirement of inertia and damping, these cycle magnitudes grow with decreasing inertia; preserving damping reverses the trend.

The central engineering implication is simple: two interventions that are individually beneficial can form a detrimental portfolio. In the demonstrated IEEE 39-bus mechanism case, a line reinforcement and controller retuning improve damping separately but reduce the damping margin by 10.76% when applied together. The proposed intervention-cycle framework predicts where such non-additivity is strongest and therefore provides a principled way to decide when isolated screening is safe and when topology and converter control must be optimized jointly.

APPENDIX A DERIVATION OF THE CONTROLLER-AWARE POLE SENSITIVITY

Starting from $T(s(\rho), \rho)v(\rho) = 0$, differentiation gives

$$(T_s s_\rho + T_\rho)v + T v_\rho = 0. \quad (65)$$

Premultiplication by a left eigenvector y^* satisfying $y^*T = 0$ removes v_ρ and gives (20). For $T = s^2M + s\Sigma + L$,

$$T_s = 2sM + \Sigma + s\Sigma_s, \quad (66)$$

$$T_\rho = s^2M_\rho + s\Sigma_\rho + L_\rho, \quad (67)$$

which yields (21). Equation (22) follows by differentiating $\zeta = -\sigma/(\sigma^2 + \omega^2)^{1/2}$.

APPENDIX B PAIR AND TRIPLE ACTION INTERACTIONS

For two rank-one actions, K has the form

$$K = \begin{bmatrix} k_{ee} & k_{ef} \\ k_{fe} & k_{ff} \end{bmatrix}. \quad (68)$$

Then

$$\det(I + K) = (1 + k_{ee})(1 + k_{ff}) - k_{ef}k_{fe}. \quad (69)$$

Thus the exact pair interaction is the feedback product $k_{ef}k_{fe}$. For three actions, the determinant additionally contains oriented products $k_{ef}k_{fg}k_{ge}$ and $k_{eg}k_{gf}k_{fe}$, consistent with the $\text{tr } Q^3$ term of (40).

APPENDIX C INDUSTRIAL DECISION EXAMPLE IN PLAIN TERMS

A utility considering a transmission reinforcement and a GFM/controller retuning might evaluate each investment independently. In the representative experiment, that procedure predicts two benefits: +1.45% and +5.39% in damping margin. The combined project, however, moves the network and controller modes into stronger hybridization and yields −10.76% relative to baseline. An interaction-aware workflow therefore avoids approving the combined portfolio without a joint study. The financial interpretation is not that the mathematical damping percentages are themselves monetary values; rather, they identify when capital projects that are individually justified by dynamic-security screens require a coupled engineering study before deployment.

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